# Bishnu Prasad Kar

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## ABOUT

AI/ML engineer and Computer Science undergraduate who builds complete systems and measures whether they work: OCR and retrieval pipelines with reported accuracy, LLM output gated by validators and red-team tests, and simulations checked against real observations. Works in Python, Rust and TypeScript, from model training to APIs and GPU code.

## EDUCATION

**B.Tech in Computer Science and Engineering** Expected Aug 2027
*Gandhi Institute of Excellent Technocrats (GIET), Bhubaneswar* *CGPA 8.8/10 (through semester 6)*
Coursework: Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Operating Systems, DBMS

## EXPERIENCE

**Data Science Intern** Dec 2025 – Feb 2026
*Jyesta Corporate Entity × E-Cell, IIT Roorkee*
- Built loan-prediction and recommendation systems: feature engineering, several classifiers, model explainability.

**Generative AI and Data Analytics Trainee** 2025 – 2026
*Central Tool Room & Training Centre (CTTC), Ministry of MSME* *Bhubaneswar*
- Generative AI (2026: LLMs, LangChain, RAG, end-to-end capstone); Data Analytics (2025: Python, SQL, Tableau).

## PROJECTS

**Nabz – Lab Reports Explained** | *Python, FastAPI, PostgreSQL, pgvector, React, TypeScript, Three.js, Docker* 2026
- Full-stack app that reads a photo or PDF of a blood-test report and explains it in three languages on a 3D body map.
- Extraction pipeline (text layer, RapidOCR with deskew, layout-agnostic parser, LOINC mapping over 108 tests): 100% field accuracy on 2,228 digital rows; 94.6% recall at 100% precision on simulated phone photos.
- Grounded LLM explanations (RAG over NLM MedlinePlus) pass a deterministic validator and an LLM judge, else fall back to a template; a 77-case red-team suite runs in CI.
- Trend engine (Theil–Sen, Mann–Kendall, RCV) confirms 88% of planted drifts at five visits, at 1.0% false alarms.

**Worldline – Relativistic Universe Sandbox** | *Rust, wgpu, egui* Oct 2026 – present
- Physics engine that simulates the real solar system from NASA JPL data (459 moons, 28,331 belt objects) with relativistic gravity and an adaptive IAS15 integrator; Earth stays within 0.41 km of JPL after a year.
- Every model ships with a validation test against theory or observation, 65+ passing: Mercury's precession 42.9807″ vs. 42.9805″ per century; a GW150914 replay leaves 63.0 $M_\odot$ (LIGO: $62 \pm 4$).
- GPU ray-traced black-hole shadows; SPH fluid planets torn apart by tides at 8,144 particles and 30+ FPS.

**Q-CONSENSUS – Multi-Agent LLM Reasoning** | *Python, FastAPI, llama.cpp, Qiskit Aer, Solidity, Docker* 2026
- Five agents debate on a local LLM; a QAOA solver over quantum-kernel similarity picks the consensus answer, benchmarked against a classical baseline.
- SHA-256 hash-chained events, anchored on a private Ethereum chain, make every run replayable and verifiable.

**Misinformation Vaccine – Fake-News Detection** | *PyTorch, Transformers, scikit-learn, FastAPI, PostgreSQL* 2025
- Fine-tuned DistilBERT on 43,944 articles to 0.9995 F1, against a 98.84%-accuracy TF-IDF baseline; documented the dataset bias behind the near-ceiling score.
- Served it as a URL-to-verdict FastAPI service with scraping and stored history.

## TECHNICAL SKILLS

**Languages**: Python, Rust, TypeScript, SQL, Bash
**Machine Learning**: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, SHAP, NLP, computer vision, OCR
**Generative AI**: LLMs, RAG, agentic systems, LangChain, FAISS, pgvector, llama.cpp, Ollama, LLM guardrails
**Systems**: FastAPI, PostgreSQL, Docker, Git, GitHub Actions, Linux, React, Spark, Hadoop, Qiskit

## PUBLICATIONS

- *Entropy-Optimized Adaptive Hybrid Thresholding for Retinal Blood Vessel Segmentation*. SPARK-2K26, 2026.
- *CityFlow: A Scalable Big Data Framework for Real-Time Analytics in Computing 4.0*. NCCENGT-2025.
- *A Modular and Scalable Arithmetic and Logical Unit Design through Multiplexer-Based Controller*. ICSTRE-2024.